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*****
MEME - Motif discovery tool
*****
MEME version 4.11.1 (Release date: Fri Jan 15 12:51:59 2016 -0800)
```

For further information on how to interpret these results or to get a copy of the MEME software please access <http://meme-suite.org> .

This file may be used as input to the MAST algorithm for searching sequence databases for matches to groups of motifs. MAST is available for interactive use and downloading at <http://meme-suite.org> .

```
*****
REFERENCE
*****
If you use this program in your research, please cite:
```

Timothy L. Bailey and Charles Elkan,  
"Fitting a mixture model by expectation maximization to discover motifs in biopolymers", Proceedings of the Second International Conference on Intelligent Systems for Molecular Biology, pp. 28-36, AAAI Press, Menlo Park, California, 1994.

```
*****
TRAINING SET
*****
DATAFILE= inpros98.txt
ALPHABET= ACDEFGHIKLMNPQRSTVWY
Sequence name      Weight Length  Sequence name      Weight Length
-----
Rosalind_7142      1.0000    240  Rosalind_4494      1.0000    240
Rosalind_3636      1.0000    240
*****
```

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*****
COMMAND LINE SUMMARY
*****
This information can also be useful in the event you wish to report a problem with the MEME software.
```

command: meme inpros98.txt -protein -oc . -nostatus -time 18000 -maxsize 60000 -mod zoops -nmotifs 1 -minw 6 -maxw 50

```

model:  mod=      zoops      nmotifs=      1      evt=      inf
object function=  E-value of product of p-values
width:  minw=      6      maxw=      50
width:  wg=      11      ws=      1      endgaps=      yes
nsites:  minsites=      2      maxsites=      3      wnsites=      0.8
theta:  spmap=      pam      spfuzz=      120
global:  substring=      yes      branching=      no      wbranch=      no
em:      prior=      megap      b=      3600      maxiter=      50
        distance=      1e-05
data:    n=      720      N=      3      shuffle=      -1

```

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sample: seed=      0      ctfrac=      -1      maxwords=      -1

```

Dirichlet mixture priors file: prior30.plib

Letter frequencies in dataset:

```

A 0.040 C 0.054 D 0.043 E 0.056 F 0.074 G 0.058 H 0.050 I 0.042 K 0.043
L 0.046 M 0.057 N 0.047 P 0.056 Q 0.058 R 0.043 S 0.047 T 0.049 V 0.036
W 0.057 Y 0.044

```

Background letter frequencies (from dataset with add-one prior applied):

```

A 0.041 C 0.054 D 0.043 E 0.055 F 0.073 G 0.058 H 0.050 I 0.042 K 0.043
L 0.046 M 0.057 N 0.047 P 0.055 Q 0.058 R 0.043 S 0.047 T 0.049 V 0.036
W 0.057 Y 0.045

```

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MOTIF 1 MEME width = 36 sites = 3 llr = 319 E-value = 9.0e-058

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# Motif 1 Description

```

Simplified      A  ::::::::::::::::::::::aa::::::::::::
pos.-specific  C  ::::a::::::::::::::::::::::::::::
probability     D  a::::::::::::::::::::::::::::a::::
matrix          E  ::::::::::::::::::::::::::::::
                F  ::::::::::::::a::::::::::::a::aa
                G  ::::::::::::::a::a::::::::::::
                H  ::::::::::a::a::::::::::::a::::a::
                I  ::::::a::::::::::::::::::::::::
                K  ::::::::::a3::::::::a::::::::::::
                L  :a::::::::::::a::::::::::::
                M  ::::::::::::::::::::::::::::::
                N  :::::::::::7::::::::::::a::

```

```

P  ::::::::::a::::a::::::::a:::::::::::::
Q  ::::::::::::::::::::::::::::::a::::::
R  ::::::::::::::::::::::a:::::::::::::
S  ::::::::::::::a::::a:::::::::::::
T  ::::::::::::::a:::::::::::::
V  ::::::::::a:::::::::::::
W  ::aa:a::::::::a::::::::a::::::::
Y  ::::::::::::::::::::::::::::::

```

```

bits      4.8      * *      **
          4.3 ***** *** *****
          3.8 ***** *****
          3.3 *****
Relative  2.9 *****
Entropy   2.4 *****
(153.4   1.9 *****
bits)    1.4 *****
          1.0 *****
          0.5 *****
          0.0 -----

```

Multilevel  
consensus  
sequence

DLWWCWIPVHKNPHSFLKTWSPAAGHRGWQFDHNFF  
K

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Motif 1 sites sorted by position p-value

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| Sequence name | Start | P-value  | Site                                                       |
|---------------|-------|----------|------------------------------------------------------------|
| Rosalind_3636 | 142   | 3.66e-47 | FHNCFRQVGM DLWWCWIPVHKNPHSFLKTWSPAAGHRGWQFDHNFF QMMGHQDWGT |
| Rosalind_7142 | 19    | 3.66e-47 | TSNMAQCRVE DLWWCWIPVHKNPHSFLKTWSPAAGHRGWQFDHNFF VYMMGFYMT  |
| Rosalind_4494 | 164   | 1.02e-46 | IKSFNKQYEH DLWWCWIPVHKNPHSFLKTWSPAAGHRGWQFDHNFF STKCSCIMSN |

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Motif 1 block diagrams

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| SEQUENCE NAME | POSITION | P-VALUE | MOTIF DIAGRAM |
|---------------|----------|---------|---------------|
| Rosalind_3636 |          | 3.7e-47 | 141_[1]_63    |

Rosalind\_7142 3.7e-47 18\_[1]\_186  
 Rosalind\_4494 1e-46 163\_[1]\_41

-----  
 Motif 1 in BLOCKS format

-----  
 BL MOTIF 1 width=36 seqs=3  
 Rosalind\_3636 ( 142) DLWWCWIPVHKNPHSFLKTWSPAAGHRGWQFDHNFF 1  
 Rosalind\_7142 ( 19) DLWWCWIPVHKNPHSFLKTWSPAAGHRGWQFDHNFF 1  
 Rosalind\_4494 ( 164) DLWWCWIPVHKKPHSFLKTWSPAAGHRGWQFDHNFF 1  
 //

-----  
 Motif 1 position-specific scoring matrix

-----  
 log-odds matrix: alength= 20 w= 36 n= 615 bayes= 8.12177 E= 9.0e-058  
 -217 -504 431 -65 -480 -318 -343 -345 -304 -319 -482 -73 -442 -357 -328 -237 -319 -284 -581  
 -392  
 -188 -397 -401 -347 -192 -392 -407 -3 -293 412 -101 -377 -371 -323 -277 -284 -239 -21 -457  
 -286  
 -299 -470 -366 -387 -202 -380 -439 -338 -339 -147 -394 -373 -450 -394 -314 -337 -357 -259 397  
 -182  
 -299 -470 -366 -387 -202 -380 -439 -338 -339 -147 -394 -373 -450 -394 -314 -337 -357 -259 397  
 -182  
 -279 415 -489 -470 -549 -501 -564 -366 -484 -371 -506 -487 -546 -546 -446 -393 -359 -370 -739  
 -565  
 -299 -470 -366 -387 -202 -380 -439 -338 -339 -147 -394 -373 -450 -394 -314 -337 -357 -259 397  
 -182  
 -171 -401 -355 -361 -292 -388 -473 405 -295 66 -169 -344 -429 -399 -339 -265 -207 182 -529  
 -312  
 -45 -427 -235 -231 -409 -257 -348 -266 -187 -193 -401 -277 389 -264 -224 -122 -184 -185 -575  
 -395  
 -13 -299 -315 -299 -316 -329 -385 88 -274 -50 -259 -335 -336 -369 -267 -227 -122 435 -546  
 -369  
 -215 -413 -197 -253 -276 -300 396 -319 -244 -193 -348 -82 -337 -95 -127 -166 -223 -238 -414  
 -81  
 -244 -498 -394 -354 -580 -406 -427 -334 437 -341 -480 -319 -435 -362 -4 -332 -323 -344 -592  
 -471  
 -124 -430 -66 -133 -401 -195 -166 -240 137 -216 -364 376 -295 -158 -79 -57 -136 -201 -495  
 -297  
 -45 -427 -235 -231 -409 -257 -348 -266 -187 -193 -401 -277 389 -264 -224 -122 -184 -185 -575

|      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| -395 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -215 | -413 | -197 | -253 | -276 | -300 | 396  | -319 | -244 | -193 | -348 | -82  | -337 | -95  | -127 | -166 | -223 | -238 | -414 |
| -81  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -24  | -308 | -226 | -278 | -393 | -214 | -362 | -276 | -204 | -239 | -377 | -144 | -294 | -311 | -224 | 397  | 97   | -212 | -531 |
| -332 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -230 | -375 | -388 | -399 | 354  | -408 | -425 | -168 | -374 | -43  | -297 | -398 | -399 | -462 | -395 | -257 | -349 | -159 | -360 |
| -67  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -188 | -397 | -401 | -347 | -192 | -392 | -407 | -3   | -293 | 412  | -101 | -377 | -371 | -323 | -277 | -284 | -239 | -21  | -457 |
| -286 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -244 | -498 | -394 | -354 | -580 | -406 | -427 | -334 | 437  | -341 | -480 | -319 | -435 | -362 | -4   | -332 | -323 | -344 | -592 |
| -471 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -72  | -327 | -249 | -282 | -381 | -270 | -359 | -158 | -197 | -196 | -298 | -153 | -324 | -275 | -223 | 64   | 395  | -87  | -518 |
| -356 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -299 | -470 | -366 | -387 | -202 | -380 | -439 | -338 | -339 | -147 | -394 | -373 | -450 | -394 | -314 | -337 | -357 | -259 | 397  |
| -182 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -24  | -308 | -226 | -278 | -393 | -214 | -362 | -276 | -204 | -239 | -377 | -144 | -294 | -311 | -224 | 397  | 97   | -212 | -531 |
| -332 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -45  | -427 | -235 | -231 | -409 | -257 | -348 | -266 | -187 | -193 | -401 | -277 | 389  | -264 | -224 | -122 | -184 | -185 | -575 |
| -395 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 423  | -220 | -277 | -254 | -358 | -115 | -389 | -189 | -236 | -156 | -308 | -272 | -335 | -320 | -251 | 6    | -125 | -23  | -503 |
| -352 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 423  | -220 | -277 | -254 | -358 | -115 | -389 | -189 | -236 | -156 | -308 | -272 | -335 | -320 | -251 | 6    | -125 | -23  | -503 |
| -352 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -94  | -458 | -212 | -294 | -499 | 392  | -401 | -375 | -252 | -357 | -475 | -212 | -381 | -386 | -264 | -166 | -297 | -285 | -550 |
| -414 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -215 | -413 | -197 | -253 | -276 | -300 | 396  | -319 | -244 | -193 | -348 | -82  | -337 | -95  | -127 | -166 | -223 | -238 | -414 |
| -81  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -238 | -418 | -375 | -380 | -517 | -357 | -283 | -337 | -57  | -277 | -466 | -298 | -357 | -255 | 436  | -280 | -319 | -351 | -485 |
| -419 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -94  | -458 | -212 | -294 | -499 | 392  | -401 | -375 | -252 | -357 | -475 | -212 | -381 | -386 | -264 | -166 | -297 | -285 | -550 |
| -414 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -299 | -470 | -366 | -387 | -202 | -380 | -439 | -338 | -339 | -147 | -394 | -373 | -450 | -394 | -314 | -337 | -357 | -259 | 397  |
| -182 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -151 | -440 | -274 | -84  | -457 | -354 | -114 | -291 | -175 | -161 | -273 | -217 | -316 | 380  | -150 | -205 | -243 | -241 | -487 |
| -389 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -230 | -375 | -388 | -399 | 354  | -408 | -425 | -168 | -374 | -43  | -297 | -398 | -399 | -462 | -395 | -257 | -349 | -159 | -360 |
| -67  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -217 | -504 | 431  | -65  | -480 | -318 | -343 | -345 | -304 | -319 | -482 | -73  | -442 | -357 | -328 | -237 | -319 | -284 | -581 |
| -392 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -215 | -413 | -197 | -253 | -276 | -300 | 396  | -319 | -244 | -193 | -348 | -82  | -337 | -95  | -127 | -166 | -223 | -238 | -414 |
| -81  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -238 | -452 | -146 | -321 | -431 | -283 | -185 | -264 | -229 | -291 | -435 | 417  | -372 | -286 | -280 | -100 | -205 | -264 | -515 |
| -358 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| -230 | -375 | -388 | -399 | 354  | -408 | -425 | -168 | -374 | -43  | -297 | -398 | -399 | -462 | -395 | -257 | -349 | -159 | -360 |

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[illegible]

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Time 0.40 secs.

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# SUMMARY OF MOTIFS

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Combined block diagrams: non-overlapping sites with p-value < 0.0001  
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| SEQUENCE NAME | COMBINED P-VALUE | MOTIF DIAGRAM        |
|---------------|------------------|----------------------|
| -----         | -----            | -----                |
| Rosalind_7142 | 7.51e-45         | 18_[1(3.66e-47)]_186 |
| Rosalind_4494 | 2.09e-44         | 163_[1(1.02e-46)]_41 |
| Rosalind_3636 | 7.51e-45         | 141_[1(3.66e-47)]_63 |

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Stopped because requested number of motifs (1) found.

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CPU: node166

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